

Glow in the Dark Skateboard: a Futuristic and Smart Skateboard

Hanna Suzuki

Bedford High School
Bedford, MA, 01730, USA

Abstract

Glow in the Dark Skateboard transforms a traditional skateboard to be futuristic and smart with a small (credit card sized) computer called Raspberry Pi and its peripherals such as an accelerometer, an electroluminescent wire and LED lights. It continuously senses the orientation of the skateboard and changes its lighting pattern according to the current orientation. This chapter describes the features of Glow in the Dark Skateboard and explains how to reproduce it with detailed instructions.

X.1 Introduction

Glow in the Dark Skateboard is a “smart” skateboard that can make skateboarding more fun and safer. It features electroluminescent wire and LED lights that flash brightly. The lights provide an amusing sight of an almost futuristic looking skateboard. It also provides safety precautions. Skateboarding can be dangerous in the dark, especially when there are cars, bikes and pedestrians around. The bright lights can alert other road users of skateboarders when it is hard to see at night.

Glow in the Dark Skateboard changes its lighting pattern in a unique way – based on the skateboard’s orientation. When the skateboard is level, LED lights fade in and out between two colors (for example, orange and its complementary color, blue). When the skateboard’s orientation is upwards or downwards, it flashes rainbow lights. This is all implemented with Python code that continuously runs on a Raspberry Pi computer to calculate the pitch angle of the skateboard with an accelerometer and flash the LED lights according to the pitch angle.

This chapter guides you to reproduce Glow in the Dark Skateboard by sharing how to assemble hardware components and how to set up software code. Potential future directions and ideas are also given to extend/customize Glow in the Dark Skateboard.

X.2 Hardware Assembly

Glow in the Dark Skateboard is built with a small (credit card sized) computer called Raspberry Pi and its peripheral components. This section explains what components to prepare and how to assemble them.

X.2.1 Required Components

The following components are required to build Glow in the Dark Skateboard.

- Raspberry Pi Zero 2 WH (1x): Purchase a MicroSD card separately. It does not come with a Raspberry Pi. Amazon Standard Identification Number (ASIN): B09LTDQY2Z. This project uses the “Version 2” of Raspberry Pi Zero (Zero 2 WH), but its Version 1 works too (Zero WH; ASIN: B0CG99MR5W).
- Flirc Aluminum Case for Raspberry Pi Zero (1x): It is highly recommended to place a Raspberry Pi in a case, so it can be firmly attached to the skateboard. ASIN: B08837L144. Adafruit product ID: 4822.
- Portable charger (1x): Any compact and thin charger works. ASIN: B0BYN7357K.
- Micro USB cable (1x): This is to connect the portable charger to Raspberry Pi. A 1-foot (30 cm) cable is recommended. ASIN: B07G934SJ9.
- Pimoroni Blink! (1x): This is a bar of 8 LEDs. It fits with Raspberry Pi’s 2x20 GPIO header. Adafruit Product ID: 3195. Pimoroni’s product Web page: [1].
- Adafruit LIS3DH Accelerometer (1x): This is used to sense the pitch angle of the skateboard. ASIN: B01BU70B64. Adafruit Product ID: 2809.
- JST SH 4-pin 1.0mm-pitch cable (1x): This type of cable is often called Adafruit STEMMA QT or SparkFun Qwiic. They are all compatible. Choose a cable with a JST-SH 4-pin connector on one end and a female socket (for GPIO pins) on the other end. It is used to connect an accelerometer with Raspberry Pi. ASIN: B09PG9MRGX. Adafruit Product ID: 4397.
- GPIO splitter (1x): This is to split Raspberry Pi’s GPIO header to two separate headers: one for an LED bar and the other for an accelerometer. ASIN: B0888W3XN4.
- 6-pin right-angle GPIO header (1x): This is convenient to wire an accelerometer and Raspberry Pi. ASIN: B00OE8GTQ8.

- 2"x10" Mounting plate (1x): This is used to mount an accelerometer. Adafruit Product ID: 5780.
- M2.5 screw set (1x): These are used to mount an accelerometer onto a mounting plate. ASIN: B0BL3RW7PJ. Adafruit Product ID: 3299.
- Electroluminescent (EL) wire (1x): A 2.5-meter (8 feet) wire is recommended. Various color choices are available. Adafruit Product ID: 405.
- EL Wire 2xAA inverter (1x): This is to supply power to an EL wire. Adafruit Product ID: 317.
- Velcro strips (4x): These are used to attach a Raspberry Pi, mounting plate, EL wire inverter and portable charger to the skateboard. ASIN: B0010HADEA.

ASINs can be used for product searches at <https://www.amazon.com>, and Adafruit product IDs can be used at <https://www.adafruit.com>.

Fig. 1 shows how Glow in the Dark Skateboard is assembled with the required components.

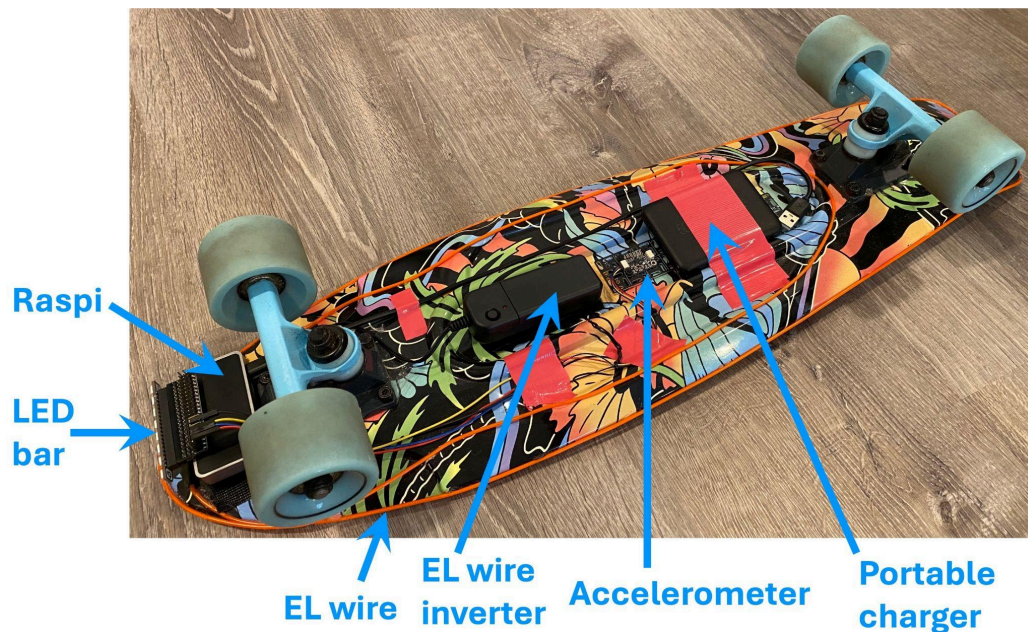


Fig. 1: Fully Assembled Glow in the Dark Skateboard

X.2.2 Attaching an Electroluminescent Wire to the Skateboard

First, the EL wire is attached onto the rim of the skateboard with hot glue (Figs. 1 and 2). By attaching it to the rim, it creates the effect that it looks floating like a hoverboard (Fig. 3). However, the wire does not have to be on the rim, it could essentially be put anywhere as long as it does not disrupt the wheels or the top of the deck. The hot glue is secure enough and dries quickly, so a lot does not have to be used. The particular wire and skateboard used had enough to wrap the perimeter of the skateboard, plus a little extra, so the extra is just glued the rest to the middle of the bottom side of the skateboard (Fig. 1).



Fig. 2: EL Wire glued on the Rim of the Skateboard



Fig. 3: EL Wire turned on in the Dark

Note that an EL wire needs to be powered by an AC power source, which a Raspberry Pi does not have. As a result, Glow in the Dark Skateboard uses an inverter (DC-AC converter) that uses two AA batteries to supply AC power for the EL wire. The EL wire is turned on and off manually with a switch on the inverter box, rather than programmatically with Raspberry Pi. The inverter box is attached to the skateboard with a velcro strip (Fig. 1).

X.2.2 Connecting an LED Bar to a Raspberry Pi

Glow in the Dark Skateboard uses Pimoroni Blinkt!, which is a bar of eight LED lights. Each light is individually controllable and dimmable. Blinkt! has 2x20 GPIO sockets (holes); it can sit directly on top of Raspberry Pi's GPIO header.

Although it sits on the GPIO header, Blinkt! actually uses three GPIO pins only [2]:

- 5V power pin (physical pin #4)
- GPIO 23 (physical pin #16)
- GPIO 24 (physical pin #18)

Therefore, you can access the rest of the GPIO pins at the same time as using Blinkt! by splitting Raspberry Pi's GPIO header to two separate headers. Connect a GPIO splitter to Raspberry Pi's GPIO header, and plug Blinkt! into one of the two headers (Figs. 4 and 5). Make sure to plug in Blinkt! in the correct way [1]. The other header will be used in the next section to wire an accelerometer.

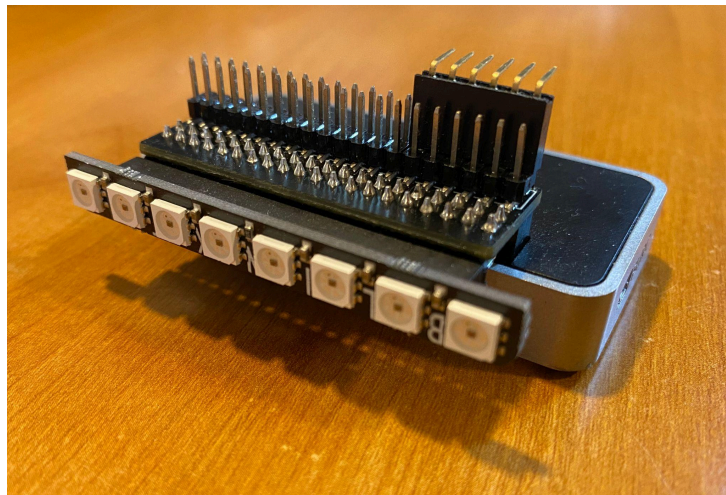


Fig. 4: LED Bar and 6-pin Right-angle Header attached to two GPIO Headers Split from Raspberry Pi (front view)



Fig. 5: LED Bar and 6-pin Right-angle Header attached to two GPIO Headers Split from Raspberry Pi (back view)

Attach Raspberry Pi to the head of the skateboard with a velcro strip (Fig. 1). Attach a portable charger to the skateboard with velcro as well, and power Raspberry Pi with a Micro USB cable (Fig. 1).

X.2.3 Connecting an Accelerometer to a Raspberry Pi

Glow in the Dark Skateboard uses the LIS3DH accelerometer. Plug a JST SH (or STEMMA QT) cable to the left connector of LIS3DH, which is circled in red in Fig. 6.

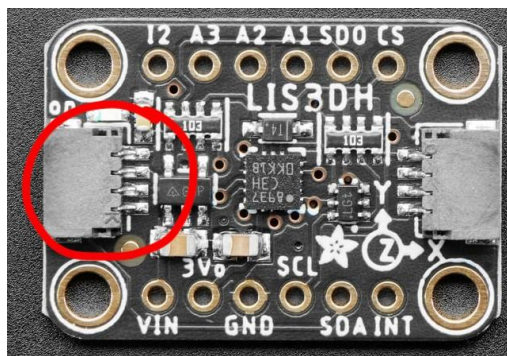


Fig. 6: LIS3DH Accelerometer

Plug a 6-pin right-angle GPIO header into the (top left) 3.3V, GPIO 2 (SDA1), GPIO 3 (SCL1), GPIO 4, GND and GPIO 7 pins (Fig. 5). Then, wire an accelerometer to 4 of them as shown in Fig. 7.

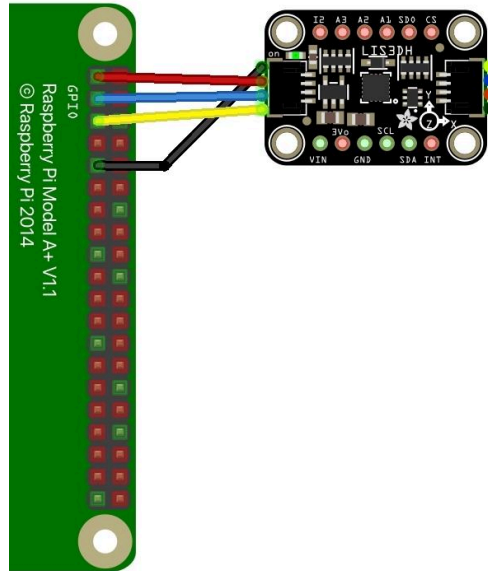


Fig. 7: Wiring of LIS3DH Accelerometer

Screw the accelerometer onto a mounting plate, and attach the plate to the skateboard with a velcro strip (Fig. 1). Make sure that the accelerometer's left connector points to the head of the skateboard. In LIS3DH, the X, Y and X axes are set up as shown in Fig. 8. Glow in the Dark Skateboard assumes the X axis faces the direction that the skateboard goes.

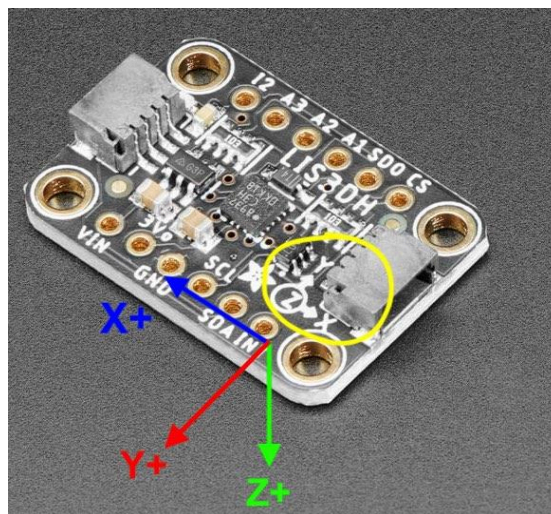


Fig. 8: Sensing Axes of LIS3DH Accelerometer

X.3 Software Development

Glow in the Dark Skateboard runs Python code in Raspberry Pi to continuously sense the orientation (pitch angle) of the skateboard and flash LED lights. This section describes how to set up and run the Python code with the Blinkt! LED bar and LIS3DH accelerometer.

X.3.1 Installing Drivers for LED Bar and Accelerometer

This section explains how to make Blinkt! and LIS3DH accessible from Python. First, turn on a Raspberry Pi and update its operating system by running the following commands one by one on a Terminal.

- `sudo apt update -y`
- `sudo apt full-upgrade -y`

Then, run the following command to make Blinkt! ready to be used.

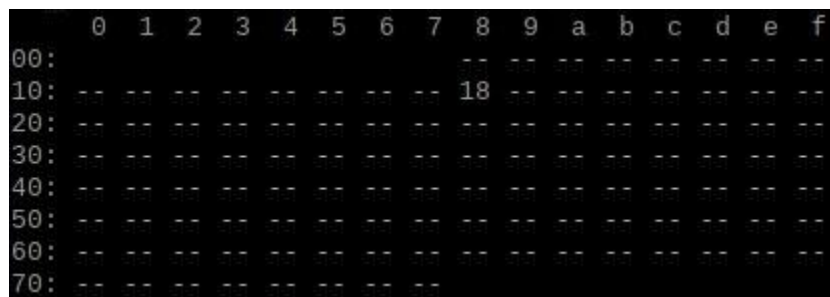
- `curl https://get.pimoroni.com/blinkt | bash`

See [3] for the reference manual of Blinkt!

In order to make LIS3DH ready to be used, enable I2C communication first with the Raspberry Pi Configuration settings. Then, run the following command on a Terminal.

- `i2cdetect -y 1`

LIS3DH is ready to communicate with Raspberry Pi through I2C, if you see “18” in the Terminal as shown in Fig. 9.



```
    0  1  2  3  4  5  6  7  8  9  a  b  c  d  e  f
00:                -- -- -- -- -- --
10: -- -- -- -- -- -- 18 -- -- -- -- --
20: -- -- -- -- -- -- -- -- -- -- -- --
30: -- -- -- -- -- -- -- -- -- -- -- --
40: -- -- -- -- -- -- -- -- -- -- -- --
50: -- -- -- -- -- -- -- -- -- -- -- --
60: -- -- -- -- -- -- -- -- -- -- -- --
70: -- -- -- -- -- -- -- -- -- -- -- --
```

Fig. 9: Output of the i2cdetect Command

The next step is to download lis3dh.py from the “code” folder at [4] and run it when the skateboard is level. You can confirm that LIS3DH is running properly if you see an output like this.

```
LIS3DH configured with addr=0x18, sensing scale=±2g.  
g: (0.052734375, 0.0390625, 0.9892578125)  
m/s^2: (0.51711328125, 0.38304687499999995, 9.815576171875)  
Pitch (degrees): -2.621751414971051 Roll (degrees): 3.01572138607504
```

Make sure that the reported pitch angle is close enough to 0, which means the skateboard is detected to be level. If the pitch is far from 0, double check if the accelerometer is attached to the skateboard as instructed earlier.

X. 3.2 Python Code

Glow in the Dark Skateboard runs the following Python code, which is available as orange-teal-fade-rainbow.py in the “code” folder at [4]. Make sure to place this code and lis3dh.py in the same folder.

```
import blinkt, time, math  
from colorzero import Color, Hue  
from lis3dh import LIS3DH  
  
sensor = LIS3DH()  
orange = Color(255, 0, 0) + Hue(deg=10)  
teal = orange + Hue(deg=180)  
red = Color(255, 0, 0)  
  
def fadeInOut(color):  
    x = 0  
    while x < 1:  
        blinkt.set_all(color.rgb_bytes[0],  
                        color.rgb_bytes[1],  
                        color.rgb_bytes[2], x)  
        blinkt.show()  
        time.sleep(0.01)  
        x = x + 0.1  
    x = 1  
    while x > 0:  
        blinkt.set_all(color.rgb_bytes[0],  
                        color.rgb_bytes[1],  
                        color.rgb_bytes[2], x)  
        blinkt.show()  
        time.sleep(0.01)  
        x = x - 0.1  
    x = 0  
  
def rainbowBlink(color):  
    i = 0  
    while i <= 7 :  
        blinkt.set_pixel(i, color.rgb_bytes[0],  
                          color.rgb_bytes[1],  
                          color.rgb_bytes[2], 1)
```

```

        color = color + Hue(deg=45)
        i = i + 1
    for x in range(5):
        blinkt.show()
        time.sleep(0.1)
        blinkt.clear()

def findPitch():
    x, y, z = sensor.readG()
    pitch, roll = sensor.pitchRoll(x, y, -z)
    pitch = math.degrees(pitch)
    return pitch

while True:
    try:
        pitch = findPitch()
        if pitch >= 5 or pitch <= -5:
            rainbowBlink(red)
        else:
            fadeInOut(orange)
            fadeInOut(teal)
    except KeyboardInterrupt:
        break
blinkt.clear()
blinkt.show()

```

X.3.3 Calculating the Pitch Angle of the Skateboard

Glow in the Dark Skateboard senses its orientation (e.g. level, upward or downward) to determine the LED bar's lighting pattern. It calculates the pitch angle as its orientation. The pitch angle is the angle between the horizontal ground surface and the base of the skateboard (θ in Fig. 10). Note that the X axis faces the direction that the skateboard goes. Therefore, when the pitch angle is close to zero, the skateboard is level. When it is positive, the skateboard is upward. It is downward when it is negative.

The pitch angle is calculated with a formula that describes the relationship between the angle (θ) and the x-component of the gravitational acceleration. In Fig. 10, the gravitational acceleration is denoted by g , and its x-component is denoted by g_x . The formula is derived as follows. Looking at Fig. 10, the pitch angle θ is congruent to the vertical angle and the angle of the right triangle with the leg g_x . The right triangle, as shown at the left bottom of Fig. 10, has the hypotenuse of g and the leg g_x . Inside of this triangle, a similar triangle can be drawn with the hypotenuse being 1 and the leg being g_x' . As a result, $g_x' = \sin \theta$. g_x' is negated in the formula because the accelerometer reports g_x as a negative value.

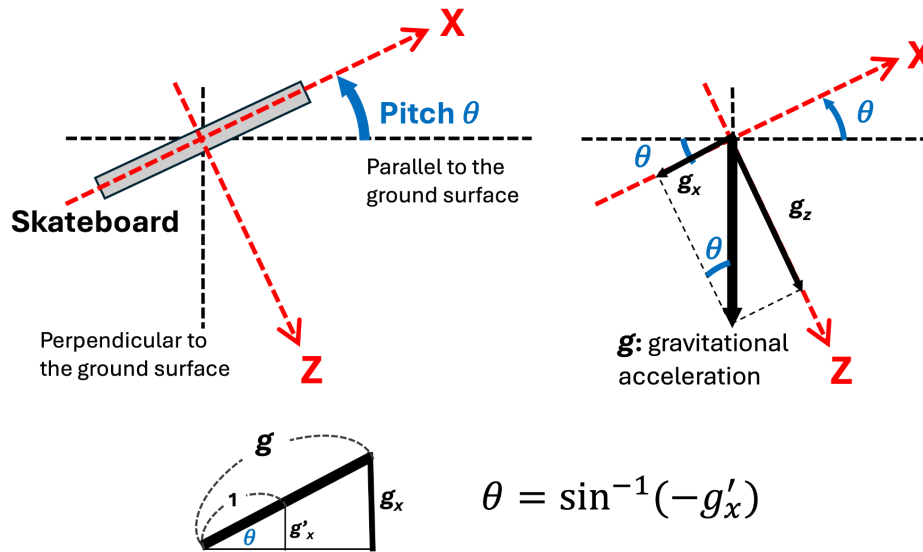


Fig. 10: Calculation of the Pitch Angle

In the Python code, the function `findPitch()` returns the pitch of the skateboard. In `findPitch()`, x-, y- and z-components of the gravitational acceleration are measured with the function `readG()` of `LIS3DH`. Then, the pitch angle is calculated with the function `pitchRoll()` of `LIS3DH` and converted from radians to degrees with the function `math.degrees()`.

X.3.4 Finding Complementary Colors and Rainbow Colors

Glow in the Dark Skateboard fades in and out of orange and its complementary color, teal blue, on the LED bar when its orientation is level. Orange is chosen because it matches the EL wire's color. You can change this choice if you use a different color for the EL wire.

Complementary colors are a pair of colors that work together by making the other brighter. When placed next to each other, they create the strongest contrast for the two colors and look vibrant together. They are on opposite sides of a color wheel, which organizes color hues around a circle and shows various relationships among them (Fig. 11). When the red color (R: 255, G: 0, B: 0) is placed at 0 degree (east), its complementary color, light blue, is at 180 degrees (west).

Glow in the Dark Skateboard uses the `colorzero` module [5] to pick up and adjust colors for LED lights. In the Python code, the following part sets up an orange color at 10 degrees counterclockwise from the red color (R: 255, G: 0, B: 0). Then, its complementary color is determined at 180 degrees from the orange color. It results in teal blue.

```
orange = Color(255, 0, 0) + Hue(deg=10)
teal = orange + Hue(deg=180)
```

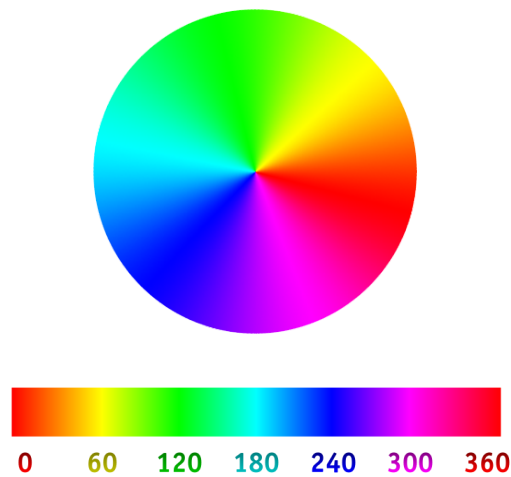


Fig. 11: Color Wheel

Glow in the Dark Skateboard also flashes rainbow colors on the LED bar when its orientation is upward or downward. Since the LED bar contains 8 lights, the rainbow colors are determined by equally dividing the color wheel into 8 parts. Since 360 degrees divided by 8 is 45, each color is 45 degrees away from each other. Starting at 0 degree, which is red, the next color is orange at 45 degrees, the third color is yellow at 90 degrees, etc. In the Python code, the function **rainbowBlink()** sets those 8 colors for the 8 LED lights and blinks them 5 times.

Currently, Glow in the Dark Skateboard considers that it is level when its pitch angle is in between -5 degrees and 5 degrees. It considers it is upward or downward when its pitch is higher than 5 degrees or lower than -5 degrees. These settings can be changed in an infinite loop at the bottom of the code.

X. 4 Future Directions

Glow in the Dark Skateboard can be extended in many different ways. One route is to focus on the health/fitness aspect of skateboarding. It is possible to detect when the skateboard moves by sensing acceleration over the x-axis because it indicates the force that a skateboarder applies to the skateboard with his/her foot. Therefore, the duration of a skateboarding session can be estimated. With extra inputs such as the skateboarder's weight and a skateboarding MET (metabolic equivalent) value, it is possible to estimate how many calories the rider has burned in the session. One MET is defined as 1 kcal/kg/hour and is roughly equivalent to the energy

expenditure of sitting quietly. MET values of 5.0 and 8.0 are estimated for general (moderate-effort) and competitive (vigorous-effort) skateboarding, respectively [6, 7, 8].

To add on to the safety aspect, a motion sensor and an ultrasonic distance sensor can be added to the skateboard. They can monitor the surrounding condition to detect other road users (e.g. cars, bikers and pedestrians) and help avoid sudden curbs or bumps on the sidewalk. For safety precaution, the sensors can trigger a sound or visual warning to alert the skateboarder of the potential safety hazard.

X. 5 Conclusion

This project delivers a technology hack to make skateboarding more fun and safer. It turns a traditional skateboard to be futuristic and smart with a Raspberry Pi computer and its peripheral components. Python code runs on the Raspberry Pi to control those peripherals. You can find the code and demo videos at [4].

References

[1] <https://shop.pimoroni.com/products/blinkt>

[2] <https://pinout.xyz/pinout/blinkt>

[3] <http://docs.pimoroni.com/blinkt/>

[4] <https://github.com/HSSBoston/skateboard>

[5] <https://colorzero.readthedocs.io/>

[6] <https://pacompendium.com/sports/>

[7] Ainsworth BE, Herrmann SD, Jacobs Jr. DR, Whitt-Glover MC, Tudor-Locke C. A brief history of the Compendium of Physical Activities. *Journal of Sport and Health Science*, 2024;13(1): 3-5.

[8] Herrmann SD, Willis EA, Ainsworth BE, Barreira TV, Hastert M, Kracht CL, Schuna Jr. JM, Cai Z, Quan M, Tudor-Locke C, Whitt-Glover MC, Jacobs DR. 2024 Adult Compendium of Physical Activities: A third update of the energy costs of human activities. *Journal of Sport and Health Science*, 2024;13(1): 6-12.